

PROJECT 5: Customer Review Insights & Response Generator

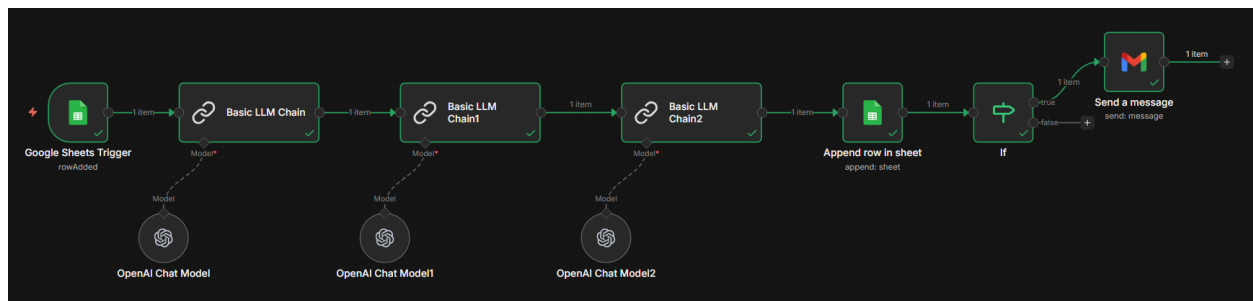
Objective

Design and deploy a multilingual, AI-powered workflow that analyzes customer reviews in real-time, generates summaries, identifies sentiment and tone, and automatically alerts relevant teams when high-risk or negative reviews are detected. The system must be fully automated using n8n, ensuring consistent evaluation, faster response times, trend visibility, and proactive customer engagement across e-commerce and SaaS platforms.

Primary Tools

- **n8n** – automation, LLM integration, Sheets & Gmail nodes
- **LLM Platform API:**
 - OpenAI (Paid models) – <https://platform.openai.com>
 - Google Gemini (Limited access to certain models under their **Free tier**) – <https://aistudio.google.com>
- **Google Sheets** – review logging & trend tracking
- **Gmail** – automated alerting for negative reviews

Workflow



[For enlarged view, click [here](#)]

Implementation Steps

Step 1: Review Data Collection

1. Create a Google Form with below fields:
 - Product Name (text)
 - Email (email)

- Rating (number 1-5)
- Feedback (text)
- 2. Collect customer reviews from Google Form to Google Sheets.
- 3. Ensure each review submitted via form is collected in the sheet.

Why: A simple form allows internal teams to test the system without external integrations.

Step 2: Start Building the Automated Workflow in n8n

1. Refer to the videos given in the reference section at the end of this document to get familiar with the tool.
2. Start n8n either via n8n cloud, or self-hosted.
3. For n8n cloud, sign in to <https://n8n.io>.
4. Click “**New Workflow**” and give it a name, e.g., *Customer Review Insights Pipeline*.
5. Add a **Google Sheets (On row added) node** as a **Trigger Node** with below settings:
 - Setup credentials for the Google Sheets node:
Credential to connect with >> *Create new credential*
 - i. *Connect using >> OAuth2*
 - ii. If running self-hosted n8n, to get the Client ID and Client Secret follow the steps shown in this [video](#) reference.
 - iii. Once Client ID and Client Secret are added, the button will appear - *Sign in with Google*. Select it to Sign in and authenticate.
 - **Mode:** Every Minute
 - **Document:** Select your Form Response Google Sheet
 - **Sheet:** Select the specific worksheet
 - **Trigger On:** Row Added
6. Text and Execute the node.
7. Retrieve the latest customer review for downstream LLM nodes.

Why: Automation eliminates manual review collection and accelerates analysis.

Step 3: Summarization Node (Basic LLM Chain Node)

1. Add a Basic LLM Chain node with below settings:
 - **Source of Prompt:** Define below
 - **Prompt:** (First select the *Expression* option then use the below prompt)

Summarize the customer's product review below in 1–2 sentences.
Return only the summary text.

Review: *{{ drag and drop the feedback text from previous Google Sheets node }}*

2. Attach a Chat Model to this Basic LLM Chain node.
Chat Model could be from Google (eg. gemini-2.5-flash), OpenAI (eg. gpt-4o-mini), etc.
3. Configure the credentials (API Keys, etc) for your Chat Model.
4. Click on **Execute step** to test the Basic LLM Chain node.

Why: Summaries give a quick, readable overview without scanning long reviews.

Step 4: Sentiment Analysis Node

1. Add a second **Basic LLM Chain** node in n8n.
2. Rename it: **Sentiment Analysis**.
3. Include a strict-output prompt:

Identify the sentiment of the product review as Positive, Neutral, or Negative.

Return in this format only:

Sentiment: <Positive/Negative/Neutral> | Score: <0–1>

Review: *{{ drag and drop the feedback text from previous Google Sheets node }}*

4. Attach the chosen LLM model and test the node.

Why: Consistent sentiment scoring is crucial for trend analysis and alerts.

Step 5: Summary Tone Classification Node

1. Add a third **Basic LLM Chain** node in n8n.
2. Rename it: **Summary Tone Classification**.
3. Write a prompt:

Classify the tone of the following review summary into:

Informative, Persuasive, or Friendly.

Return in this format only:

Tone: <Informative/Persuasive/Friendly> | Score: <0–1>

Review Summary: *{{ drag and drop the product review summary text from summarization node }}*

4. Execute and test the node.

Why: Tone profiling helps understand how customers express experiences.

Step 6: Store Results in Google Sheets

1. Create another Google Sheets (OR create a new worksheet in the same Google Sheets where Form responses were being saved).
2. To this new worksheet, add the below column fields to store the insights:
 - Product Name
 - Product Review
 - Summary
 - Sentiment & Score
 - Tone & Score
3. Connect the above Google Sheets in your n8n workflow.
4. Add a **Google Sheets (Append row in sheet) node** with below settings:
 - Setup credentials for the Google Sheets node:
Credential to connect with >> *Create new credential*
 - i. *Connect using >> OAuth2*
 - ii. If running self-hosted n8n, to get the Client ID and Client Secret follow the steps shown in this [video](#) reference.
 - iii. Once Client ID and Client Secret are added, the button will appear - *Sign in with Google*. Select it to Sign in and authenticate.
 - **Resource:** Sheet Within Document
 - **Operation:** Append Row
 - **Document:** Select your Google Sheet
 - **Sheet:** Select the specific worksheet
 - **Mapping Column Mode:** Map Each Column Manually
 - i. **Product Name:** *“drag and drop the product name text from previous Google Sheets node”*
 - ii. **Product Review:** *“drag and drop the product review text from previous Google Sheets node”*
 - iii. **Summary:** *“drag and drop the summary text from summarization node”*
 - iv. **Sentiment & Score:** *“drag and drop the sentiment text from sentiment analysis node”*
 - v. **Tone & Score:** *“drag and drop the tone text from tone classification node”*
5. Click on **Execute step** to test and execute the node.
6. Check the insights being added to the Google Sheets as a new row.

Why: Sheets serve as a live analytics database and audit log.

Step 7: Negative Review Detection Logic

1. Add an **If Node** and rename it: **Is Review Negative?**
2. Set condition to check if the Sentiment output contains “Negative”.
3. If **True** → **Send an email alert**
4. If **False** → **Workflow ends normally**

Why: Immediate alerts for negative reviews prevent escalations and churn.

Step 8: Automated Email Alert to Team Lead

1. At the **True branch** of the If node, add a **Gmail Node**.
2. Fill in:
 - **To:** Team Lead Email
 - **Subject:** Negative Customer Review Received – {{ Product Name }}
 - **Message Body:**

Dear Team Lead,
We have received a negative customer review for: {{ Product Name }}.

Summary: {{ Product Review Summary }}

Please review and take necessary action.

Best Regards,
Workflow Automation Team

3. Test the node to confirm email delivery.

Why: Real-time notifications help teams respond proactively.

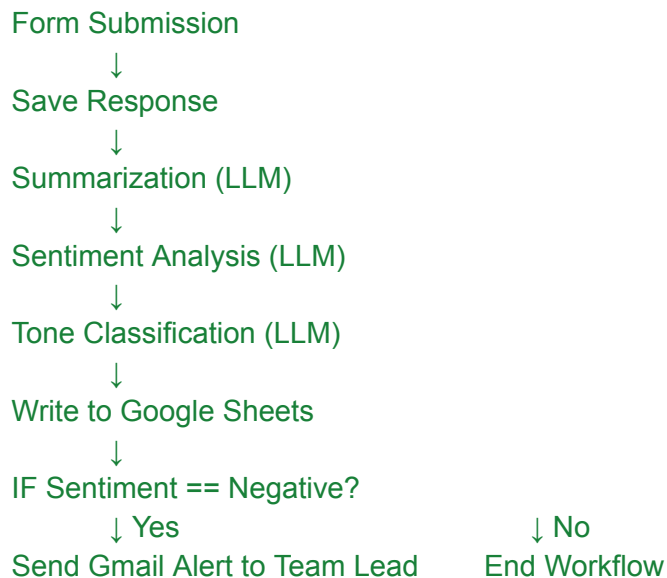
Step 9: Activate, Deploy & Test Workflow

1. Save and activate the n8n workflow.
2. Submit test reviews to validate:
 - Form → Save Response → Summary → Sentiment → Tone → Sheets → Alerts

3. Debug failed nodes if logs show errors.

Why: Full testing ensures a reliable, production-ready automation.

Workflow Schematic



Deliverables

1. n8n Workflow Export (.json) full pipeline: form → LLM agents → Sheets → conditional alerts → Gmail.
2. Google Sheet Review Log: stored summaries, sentiment, tone, historical insights and dashboards.
3. Sample Input & Output Set: Example reviews and system-generated insights.
4. Alert Email Samples: Screenshots or test messages for negative review alerts

Reference - [Challenge 4: Sentiment Analysis using Hugging Face Spaces](#)
[Challenge 8: Article Summary & Analysis using ChatGPT](#)

PROJECT 6: Enhancing Customer Service with Agent-powered Chatbots

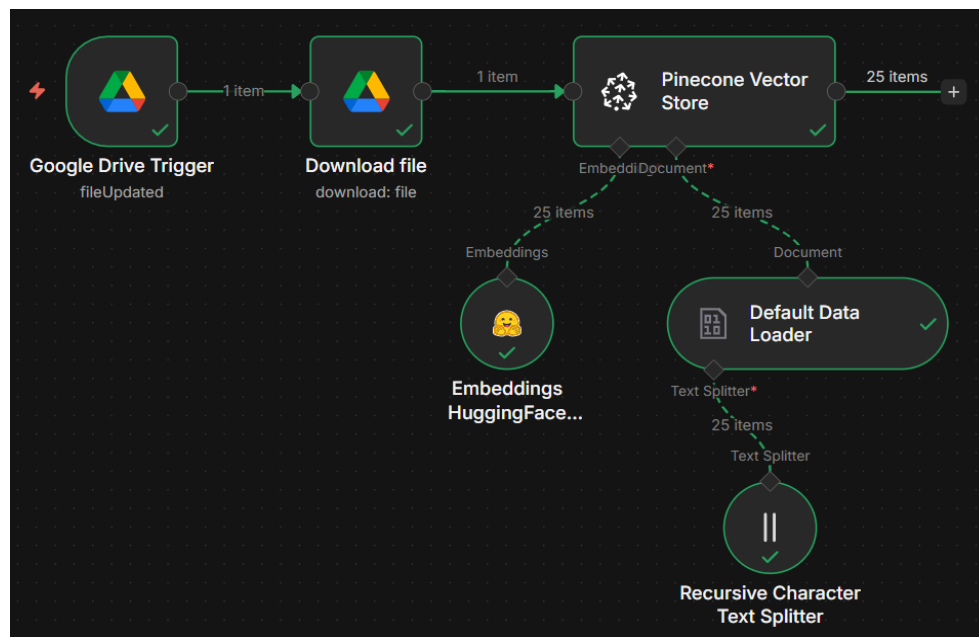
Objective

Build a RAG-based customer support assistant that allows users to ask natural-language questions and receive accurate, document-grounded answers sourced strictly from E-commerce policies. The system must use n8n as the orchestration engine and ensure responses are faithful to the documents, reducing hallucinations, support workload, and resolution time.

Primary Tools

- **n8n** (workflow orchestration) – <https://n8n.io>
- **LLM Platform API:**
 - OpenAI (Paid models) – <https://platform.openai.com>
 - Google Gemini (Limited access to certain models under their **Free tier**) – <https://aistudio.google.com>
- **Vector Databases:** Pinecone – <https://www.pinecone.io>
- **Delivery Channels:** Email, Webhook, or Chat UI

Workflow 1 and 2



[For enlarged view, click [here](#)]